latents verification matrix

Package 1.3.2. Built from saved evidence on 2026-11-14; nothing was refitted. 39,830 current checks (3,913,814 values) against references other than latents itself.

A **check** is one harness quantity (a whole output column counts once; *values* counts every number inside it), one argument-sweep check or one testthat expectation. **Current** requires matching runtime-source and result-file hashes in an evidence sidecar. Historical 1.0.0 evidence carries an explicit retrospective source-equivalence attestation, not a claimed rerun. Missing or mismatched provenance is stale and does not count toward strength.

Legend

Strength	One **+** per *source*: an independent program or package (another R package, Mplus, Latent GOLD, Stata, Python) whose results agree on the current code for a model with two or more classes. One **P** per published paper whose printed numbers are reproduced. **++++ PP** is four sources and two papers. Colour: **0** no source, **+** one, **++** two, **+++** three or more.
Not counted	Agreement on a one-class reduction only (one class equals lm(), glm.nb(), polr(), lme4, a linear mixed model), shown as "(1 class)"; component checks (formulas and helper routines); our own independent code (numDeriv, dense likelihoods, the kits' reference implementations); internal checks; stale results; sources with failed or incomplete tests.
Sources, papers	The number of + and of P.
Datasets	Distinct datasets on which the function was compared with a source or a paper on current code, for models with two or more classes: the harness's datasets, the argument sweep's datasets (each case's data= option), the kits' datasets (kit-datasets.csv) and the simulated datasets of the testthat kits that the other program ran (test-datasets.csv). A dataset used by several kits counts once. Tests that build their own small data inline are not counted, so this is a lower bound.
Checks passed	Current checks against every reference other than latents itself: one harness quantity (a whole output column), one kit quantity, one argument-sweep check or one testthat expectation. *Values* counts the numbers inside the harness checks.
Failed	Current disagreements without a registered explanation, plus test errors. Skipped or incomplete tests provide no source credit.
Grid cells	**bold**: current checks on mixtures; (1 class): one-class reductions only; *grey italic*: stale provenance or no saved result; candidate: found, not yet wired in; *: a test that checks several functions or references at once (counted once in each function's total).

Strength by function

function	strength	sources	papers	datasets	checks passed	failed	incomplete	values (harness)	independent programs and packages	published papers	to strengthen
lpa()	**++++++++ PPP**	8	3	236	15,763	0	0	223,780	Mplus, Latent GOLD, Stata 19 [SEM] Example 52g (gsem), mclust, Rmixmod, MoEClust, mixtools, EMCluster	Masyn (2013), Oxford Handbook of Quantitative Methods, Table 25.12, Scrucca et al. (2023) mclust book, Houle et al. (2026)	one-class checks only, not counted: stats lm/glm
multilpa()	**+++++**	5	0	81	8,125	0	0	361,890	Mplus, Latent GOLD, depmixS4, mclust, flexmix		one-class checks only, not counted: sandwich; no published paper found
lca()	**+++++++++ PPPPPPP**	9	7	103	6,599	0	1	283,886	Mplus, Latent GOLD, Stata 19 [SEM] Example 50g (gsem), poLCA, depmixS4, glca, BayesLCA, randomLCA, multilevLCA	Goodman (2002), Kleinman and Horton (2011), SAS and R blog, Example 8.21, Linzer & Lewis (2011), Dayton (1998), Zhou & Lange (2010), McCutcheon (1987), Agresti & Lang (1993)	1 incomplete tests
multilca()	**++++ P**	4	1	27	5,410	0	0	2,557,037	Mplus, Latent GOLD, glca, multilevLCA	Lyrvall et al. (2025)	
lta()	**+++++ PPP**	5	3	32	2,853	0	0	347,114	Mplus, Latent GOLD, depmixS4, LMest, seqHMM	Pennoni, Pandolfi and Bartolucci (2024), R Journal RJ-2024-036,	

function	strength	sources	papers	datasets	checks passed	failed	incomplete	values (harness)	independent programs and packages	published papers	to strengthen
										section 3.3, Muthén & Asparouhov (2022), Bartolucci, Pandolfi and Pennoni (2017), JSS 81(4), section 3.1	
mixture_regression()	+++++ PPP	5	3	22	1,010	0	0	87,830	Latent GOLD, Mplus, flexmix, MoEClust, mixtools	SAS/STAT 14.2 PROC FMM Example 40.3 (Wang et al. 1996 model), Benaglia et al. (2009) JSS 32(6), CO2data, Grun and Leisch (2008) JSS 28(4), betablocker	one-class checks only, not counted: stats lm/glm, MASS, sandwich
mixture_regression(random=)	+++ P	3	1	16	941	0	3	40,044	Mplus (Mplus 9 Demo (JStats fixtures); Mplus), Latent GOLD, SAS PROC TRAJ (Andruff et al. 2009)	Proust-Lima, Philipps and Liquet (2017), JSS 78(2), section 6	3 incomplete tests; one-class checks only, not counted: nlme, lme4
three_step() / r3step()	++++	4	0	82	2,854	0	0	1,617	Latent GOLD, Mplus, depmixS4, sandwich		first paper from: Bakk & Kuha (2021) BJMSP
bootstrap_lrt() / VLMR	+++	3	0	64	5,506	0	0	12	Latent GOLD, Mplus, mclust		first paper from: Masyn (2013) Table 25.12 and Stata lcstats
enumerate_*() / compare_models()	++++	4	0	8	227	0	0	156	Mplus, mclust, poLCA, glca		no published paper found
pool_imputations()	+++	3	0	2	805	0	0	10,448	mitools, mice, mitml		no published paper found

Reference by function

Each cell: the current checks of a function against a reference (styles in the legend). Internal checks (invariance, seed stability, refusal parity, identities) are in the evidence pages, not in this grid.

reference	lpa()	multilpa()	lca()	multilca()	lta()	mixture_ regression()	growth (random =)	three_step() r3step()	bootstrap_ lrt() VLMR	enumerate_* () compare_ models()	pool_ imputations()
strength (+ source, P paper)	++++++++ PPP	+++++	+++++++++ PPPPPPP	++++ P	+++++ PPP	+++++ PPP	+++ P	++++	+++	++++	+++
datasets tested	**236**	**81**	**103**	**27**	**32**	**22**	**16**	**82**	**64**	**8**	**2**
COMMERCIAL PROGRAM											
Mplus	**6,601***	**3,179***	**3,469***	**3,601***	**136**	**35**	**18**	**46**	**877**	**51**	
Latent GOLD	**4,324***	**2,446***	**808***	**649***	**134**	**120**	**91**	**539**	**1,204**		
Mplus 9 Demo (JStats fixtures)							**134**				
Stata 19 [SEM] Example 52g (gsem)	**74**										
SAS PROC TRAJ (Andruff et al. 2009)							**36**				
Stata 19 [SEM] Example 50g (gsem)			**22**								
IMMERSE lca-bookdown (Mplus 8.11)					candidate						
IMMERSE Nylund-Gibson et al. (2019)								candidate			
Mplus 9 Demo LCGA/GMM (JStats)							candidate				
Mplus Demo TECH11/TECH14 (JStats)									candidate		
Mplus UG ex10.13					candidate						
Mplus UG ex7.6/7.7/7.8 (JStats)			candidate								
Mplus Web Talk 8 BCH (JStats)								candidate			
Stata 19 [FMM] Examples 1a/1b/2						candidate					
Stata 19 [SEM] Example 52g	candidate										
Stata 19 [SEM] Examples 50g/51g			candidate								
R PACKAGE											
mclust	**2,844**	**12**							**8**	**122***	
depmixS4		**290**	**52**		**1,922***			**157**			
poLCA			**1,926***							**47***	
glca			**52**	**555**						**2***	
multilevLCA			**25**	**549**							
lcmm							incomplete				
flexmix		**4**				**507**					
LMest					**395***						
mitools											**163**
sandwich		95 (1 class)				1 (1 class)		**56**			
MoEClust	**55**					**25**					
Rmixmod	**75**										
mixtools	**50**					**20**					
mice											**62**
BayesLCA			**33**								
stats lm/glm	13 (1 class)					17 (1 class)					
EMCluster	**30**										
randomLCA			**30**								

reference	lpa()	multilpa()	lca()	multilca()	lta()	mixture_ regression()	growth (random =)	three_step() r3step()	bootstrap_ lrt() VLMR	enumerate_* () compare_ models()	pool_ imputations()
mitml											**20**
MASS						15 (1 class)					
seqHMM					**8**						
nlme							5 (1 class)				
lme4							4 (1 class)				
mclust bootstrap LRT (2016 paper; vignette)									candidate		
mclust vignette (Scrucca et al.), diabetes									*no result*		
nlme corAR1/corCAR1 identity							candidate				
randomLCA (Beath 2017)									candidate		
Rmixmod (Lebret et al. 2015)	candidate										
PYTHON PACKAGE											
StepMix (Morin et al. 2025)	candidate		candidate								
StepMix (Morin et al. 2025) Table 9								candidate			
PUBLISHED											
Masyn (2013), Oxford Handbook of Quantitative Methods, Table 25.12	**49**										
Proust-Lima, Philipps and Liquet (2017), JSS 78(2), section 6							**19**				
SAS/STAT 14.2 PROC FMM Example 40.3 (Wang et al. 1996 model)						**14**					
Goodman (2002)			**12**								
Pennoni, Pandolfi and Bartolucci (2024), R Journal RJ-2024-036, section 3.3					**12**						
Kleinman and Horton (2011), SAS and R blog, Example 8.21			**11**								
Linzer & Lewis (2011)			**9**								
Muthén & Asparouhov (2022)					**8**						
Dayton (1998)			**6**								
Zhou & Lange (2010)			**6**								
McCutcheon (1987)			**5**								
Bartolucci, Pandolfi and Pennoni (2017), JSS 81(4), section 3.1					**4**						
Benaglia et al. (2009) JSS 32(6), CO2data						**4**					
Agresti & Lang (1993)			**3**								
Lyrvall et al. (2025)				**3**							
Scrucca et al. (2023) mclust book	**3**										
Grun and Leisch (2008) JSS 28(4), betablocker						**2**					
Houle et al. (2026)	**2**										
Bakk & Kuha (2021) BJMSP								candidate			
Deb & Trivedi (1997)						candidate					
Masyn (2013)	candidate										
Masyn (2013) Table 25.12 and Stata lcstats									candidate		
Scrucca et al. (2016)	candidate										
COMPONENT CHECK											

reference	lpa()	multilpa()	lca()	multilca()	lta()	mixture_ regression()	growth (random =)	three_step() r3step()	bootstrap_ lrt() VLMR	enumerate_* () compare_ models()	pool_ imputations()
mclust parameter counts	168	1									
stats lm/glm (one-class reduction)	1	37				96					
lme4 (one-class reduction)		96 (1 class)					24 (1 class)				
stats p.adjust								42			51
mice pool.scalar											52
mclust M-step	10	39									
tidySEM classification-error matrix								43			
closed form (one-class reduction)	28	6 (1 class)	8 (1 class)								
stats lm/glm		1									
MASS glm.nb (one-class reduction)		12 (1 class)									
mclust MAP M-step	12										
tidyLPA classification formulas	11										
poLCA (one-class reduction)			8 (1 class)								
stats chisq.test			6								
mclust BIC convention	5										
mclust hypervolume	4										
sandwich (glm composition)						4					
Latent GOLD weight-convention reference									3		
stats lm (one-class reduction)	3										
stats weighted-logit M-step		3									
MASS polr (one-class reduction)						2 (1 class)					
mclust E-step	2										
poLCA parameter counts			2								
sandwich (lm composition)						1					
INDEPENDENT CODE											
VLMR independent reference									3,381		
ML three-step reference (reference.R)								1,793			
likelihood rebuilt from the result tables	664	640	74	36	103	112	32				
numDeriv / dense likelihood	148*	507*	7		135	27	49	48*	33		
base-R likelihood + numDeriv	125	321		17				18			16
direct-effects independent implementation	334										
Rubin's rules written here											294
published formula computed here	50							90			147
mvtnorm dense density		178									
custom EM using MASS		160									
custom EM using VGAM		83									
reference optimizer	22*	15	incomplete		1	3		25*			
pair-covariance independent likelihood and derivatives	65										
joint-items independent EM			14								
Kneedle hand-worked and reported values										7	
cluster bootstrap reference								3			
Rmpfr exact arithmetic						3					
hand-built CR1 sandwich						2					

reference	lpa()	multilpa()	lca()	multilca()	lta()	mixture_ regression()	growth (random =)	three_step() r3step()	bootstrap_ lrt() VLMR	enumerate_* () compare_ models()	pool_ imputations()
current external/component/oracle checks	15,763	8,125	6,599	5,410	2,853	1,010	941	2,854	5,506	227	805

Evidence by function

lpa()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Mplus	commercial program	current	6,601	212	220			*
Latent GOLD	commercial program	current	4,324	213	4,950			*
Stata 19 [SEM] Example 52g (gsem)	commercial program	current	74	1	74			
mclust	R package	current	2,844	78	89,114			
Rmixmod	R package	current	75	6	7,898			
MoEClust	R package	current	55					
mixtools	R package	current	50	6	5,022			
EMCluster	R package	current	30	6	3,128			
stats lm/glm	R package	one class only	13					
Masyn (2013), Oxford Handbook of Quantitative Methods, Table 25.12	published	current	49	1	49			
Scrucca et al. (2023) mclust book	published	current	3	1	3			
Houle et al. (2026)	published	current	2					
mclust parameter counts	component check	current	168		168			
closed form (one-class reduction)	component check	current	28		134			
mclust MAP M-step	component check	current	12					
tidyLPA classification formulas	component check	current	11					
mclust M-step	component check	current	10					
mclust BIC convention	component check	current	5					
mclust hypervolume	component check	current	4					
stats lm (one-class reduction)	component check	current	3		3			
mclust E-step	component check	current	2					
stats lm/glm (one-class reduction)	component check	current	1					
likelihood rebuilt from the result tables	independent code	current	664		105,182			
direct-effects independent implementation	independent code	current	334					
numDeriv / dense likelihood	independent code	current	148		165			*
base-R likelihood + numDeriv	independent code	current	125		3,421			
pair-covariance independent likelihood and derivatives	independent code	current	65					
published formula computed here	independent code	current	50		4,249			
reference optimizer	independent code	current	22					*
internal check	internal	current	1,889		58,034			
search protocol	internal	current	546					*
invariance	internal	current	352		15,520			
vignette claims	internal	current	15					
internal identity	internal	current	8					
refusal parity	internal	current	7		7			
seed stability	internal	current	3		3			
simulation recovery	internal	current	2					

Candidates not yet wired in

reference	kind	data	published numbers	checked
Masyn (2013)	published	Reaven-Miller diabetes /10 (Stata gsem_lca2.dta, n = 145)	Table 25.12: LL, BIC, adj. LMR for K = 1-5 x EEI/VVI/EEE/VVV (VVV-3 LL -1536.64); Table 25.14 entropy .88, AvePP, OCC	verified
Stata 19 [SEM] Example 52g	commercial program	same as Masyn (2013)	EEI-2 LL -1702.5542, VVV-3 LL -1536.6409, every parameter with SE	verified
StepMix (Morin et al. 2025)	Python package	iris; mclust::banknote	iris VVV-3 LL -180.185; banknote VVI-2 with covariate LL -771.669	verified
Scrucca et al. (2016)	published	old mclust::diabetes; gclus::wine	14 structures x G = 1-9 BIC grid; wine EVE-3 BIC -6873.257; BLRT statistics	verified
Rmixmod (Lebret et al. 2015)	R package	geyser	K = 3 LL -1131.0738, BIC 2312.5998	verified

multilpa()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Mplus	commercial program	current	3,179	56				*
Latent GOLD	commercial program	current	2,446	78	135,142	14		*
depmixS4	R package	current	290	3	13,772			
sandwich	R package	one class only	95		3,474			
mclust	R package	current	12					
flexmix	R package	current	4		4			
lme4 (one-class reduction)	component check	one class only	96		216			
mclust M-step	component check	current	39					
stats lm/glm (one-class reduction)	component check	current	37					
MASS glm.nb (one-class reduction)	component check	one class only	12		16			
closed form (one-class reduction)	component check	one class only	6		41			
stats weighted-logit M-step	component check	current	3					
mclust parameter counts	component check	current	1					
stats lm/glm	component check	current	1					
likelihood rebuilt from the result tables	independent code	current	640		177,005			
numDeriv / dense likelihood	independent code	current	507					*
base-R likelihood + numDeriv	independent code	current	321		17,495			
mvtnorm dense density	independent code	current	178					
custom EM using MASS	independent code	current	160		9,032			
custom EM using VGAM	independent code	current	83		5,693			
reference optimizer	independent code	current	15					
internal check	internal	current	1,583		77,194			
invariance	internal	current	279		10,649			
simulation recovery	internal	current	21					
internal identity	internal	current	10					
refusal parity	internal	current	10		10			
search protocol	internal	current	4					*

Candidates not yet wired in

reference	kind	data	published numbers	checked
singleton-group identity	internal	any lpa() reference	groups of size one: multilpa() equals lpa()	identity

lca()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Mplus	commercial program	current	3,469	77	411			*
Latent GOLD	commercial program	current	808	71	5,116			*
Stata 19 [SEM] Example 50g (gsem)	commercial program	current	22	1	22			
poLCA	R package	current	1,926	70	174,469			*
depmixS4	R package	current	52	9	30,914			
glca	R package	current	52	9	30,914			
BayesLCA	R package	current	33	8	305			
randomLCA	R package	current	30	7	277			
multilevLCA	R package	current	25	8	260			
Goodman (2002)	published	current	12	1	12			
Kleinman and Horton (2011), SAS and R blog, Example 8.21	published	current	11	1	11			
Linzer & Lewis (2011)	published	current	9	1	9			
Dayton (1998)	published	current	6	1	6			
Zhou & Lange (2010)	published	current	6	1	6			
McCutcheon (1987)	published	current	5	1	5			
Agresti & Lang (1993)	published	current	3	1	3			
closed form (one-class reduction)	component check	one class only	8		94			
poLCA (one-class reduction)	component check	one class only	8		8			
stats chisq.test	component check	current	6					
poLCA parameter counts	component check	current	2		2			
likelihood rebuilt from the result tables	independent code	current	74		41,042			
joint-items independent EM	independent code	current	14					
reference optimizer	independent code	incomplete	11					
numDeriv / dense likelihood	independent code	current	7					
internal check	internal	current	346		23,228			
invariance	internal	current	235		14,524			
search protocol	internal	current	44					*
verification protocol	internal	current	21					
vignette claims	internal	current	13					
refusal parity	internal	current	5		5			
seed stability	internal	current	3		3			
internal identity	internal	current	2					

Candidates not yet wired in

reference	kind	data	published numbers	checked
Stata 19 [SEM] Examples 50g/51g	commercial program	Stouffer-Toby (gsem_lca1.dta, n = 216)	2 classes LL -504.46767 with SEs; entropy and LMR for K = 1-3	verified
Mplus UG ex7.6/7.7/7.8 (JStats)	commercial program	UG data in ../JStats/validation/r-reference	ex7.6 LL -19214.480; ex7.7/7.8 LL -21408.482, estimates with SEs	verified
StepMix (Morin et al. 2025)	Python package	data_bakk_covariate(2000) exported from Python	LL -5192.049 (StepMix = poLCA = Mplus)	verified

multilca()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Mplus	commercial program	current	3,601	16	395,792			*
Latent GOLD	commercial program	current	649	18	15			*
glca	R package	current	555	4	72,424			
multilevLCA	R package	current	549	4	2,056,646			
Lyrvall et al. (2025)	published	current	3	1	3			
likelihood rebuilt from the result tables	independent code	current	36		26,004			
base-R likelihood + numDeriv	independent code	current	17		6,153			
internal check	internal	current	75		10,470			
search protocol	internal	current	4					*
seed stability	internal	current	3		3			
refusal parity	internal	current	2		2			

Candidates not yet wired in

reference	kind	data	published numbers	checked
singleton-group identity	internal	any lca() reference	groups of size one: multilca() equals lca()	identity

lta()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Mplus	commercial program	current	136	4	100			
Latent GOLD	commercial program	current	134	6	9,731			
depmixS4	R package	current	1,922	17	203,251			*
LMest	R package	current	395	5	51,031			*
seqHMM	R package	current	8	1	45			
Pennoni, Pandolfi and Bartolucci (2024), R Journal RJ-2024-036, section 3.3	published	current	12	1	12			
Muthén & Asparouhov (2022)	published	current	8					
Bartolucci, Pandolfi and Pennoni (2017), JSS 81(4), section 3.1	published	current	4	1	4			
numDeriv / dense likelihood	independent code	current	135					
likelihood rebuilt from the result tables	independent code	current	103		82,940			
reference optimizer	independent code	current	1					
internal check	internal	current	302		434			
refusal parity	internal	current	11		11			

Candidates not yet wired in

reference	kind	data	published numbers	checked
IMMERSE lca-bookdown (Mplus 8.11)	commercial program	LSAY lsay_lta.csv	invariant LL -14542.768; occasion-specific LL -14531.997	verified
Mplus UG ex10.13	commercial program	posted with the example	two-level LTA LL -20716.373 (needs group-class effects on part of the logits)	online

mixture_regression()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Latent GOLD	commercial program	current	120	7	4,918			
Mplus	commercial program	current	35	1	29			
flexmix	R package	current	507	10	31,143			
MoEClust	R package	current	25	5	1,779			
mixtools	R package	current	20	4	970			
stats lm/glm	R package	one class only	17					
MASS	R package	one class only	15					
sandwich	R package	one class only	1					
SAS/STAT 14.2 PROC FMM Example 40.3 (Wang et al. 1996 model)	published	current	14	1	14			
Benaglia et al. (2009) JSS 32(6), CO2data	published	current	4	1	4			
Grun and Leisch (2008) JSS 28(4), betablocker	published	current	2	1	2			
stats lm/glm (one-class reduction)	component check	current	96		41			
sandwich (glm composition)	component check	current	4					
MASS polr (one-class reduction)	component check	one class only	2		4			
sandwich (lm composition)	component check	current	1					
likelihood rebuilt from the result tables	independent code	current	112		48,926			
numDeriv / dense likelihood	independent code	current	27					
reference optimizer	independent code	current	3					
Rmpfr exact arithmetic	independent code	current	3					
hand-built CR1 sandwich	independent code	current	2					
internal check	internal	current	153		180			
invariance	internal	current	9		9			
refusal parity	internal	current	7		7			
internal identity	internal	current	1					

Candidates not yet wired in

reference	kind	data	published numbers	checked
Stata 19 [FMM] Examples 1a/1b/2	commercial program	mus03sub.dta, gsem_mixture.dta	3-class Gaussian LL -4727.6738; lcprob LL -4712.3871; Poisson LL -11502.686	verified
Deb & Trivedi (1997)	published	AER::NMES1988	2-class negative binomial (tables not opened)	online

mixture_regression(random =)

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Mplus 9 Demo (JStats fixtures)	commercial program	current	134	4	134			
Latent GOLD	commercial program	current	91	3	5,688			
SAS PROC TRAJ (Andruff et al. 2009)	commercial program	current	36	1	36			
Mplus	commercial program	current	18	1	18			
lcmm	R package	incomplete	529	7	34,083			
nlme	R package	one class only	5					
lme4	R package	one class only	4					
Proust-Lima, Philipps and Liquet (2017), JSS 78(2), section 6	published	current	19	1	19			
lme4 (one-class reduction)	component check	one class only	24		34			
numDeriv / dense likelihood	independent code	current	49					
likelihood rebuilt from the result tables	independent code	current	32		32			
internal check	internal	current	116		128			
unit-scale refit	internal	current	10		824			
internal identity	internal	current	9					
unit invariance	internal	current	1					

Candidates not yet wired in

reference	kind	data	published numbers	checked
Mplus 9 Demo LCGA/GMM (JStats)	commercial program	../JStats/tests/fixtures/mplus (lcga-ml; gmm-ml; runner-*)	LCGA K = 2 LL -2091.416; GMM K = 2 LL -2727.520	verified
nlme corAR1/corCAR1 identity	R package	any growth data	one class with AR/Brownian residuals equals nlme::lme	identity

three_step() / r3step()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Latent GOLD	commercial program	current	539	79	330			
Mplus	commercial program	current	46	70				
depmixS4	R package	current	157	3	638			
sandwich	R package	current	56	2	106			
tidySEM classification-error matrix	component check	current	43		93			
stats p.adjust	component check	current	42		84			
ML three-step reference (reference.R)	independent code	current	1,793					
published formula computed here	independent code	current	90		237			
numDeriv / dense likelihood	independent code	current	48		36			*
reference optimizer	independent code	current	25		33			*
base-R likelihood + numDeriv	independent code	current	18		60			
cluster bootstrap reference	independent code	current	3					
internal check	internal	current	22		34			
runtime guard	internal	current	10					
internal identity	internal	current	6					

Candidates not yet wired in

reference	kind	data	published numbers	checked
StepMix (Morin et al. 2025) Table 9	Python package	GSS 1976-77 via gssr	BCH 26.71/36.81/44.68; ML LL -18754.72	verified
Bakk & Kuha (2021) BJMSP	published	GSS 1976-77 (ICPSR)	BCH 26.65/37.06/44.70; ML 20.63/44.94/65.67	online
Mplus Web Talk 8 BCH (JStats)	commercial program	mplus-webtalk8-distal.dat	LL -3634.865; BCH means with SEs	verified
IMMERSE Nylund-Gibson et al. (2019)	commercial program	LSAY lsay_df.csv	BCH step 3 LL -12744.043; ML step 3 LL -12742.252	verified

bootstrap_lrt() / VLMR

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Latent GOLD	commercial program	current	1,204	62				
Mplus	commercial program	current	877	63	4			
mclust	R package	current	8	1	8			
mclust vignette (Scrucca et al.), diabetes	R package	no saved result	0					
Latent GOLD weight-convention reference	component check	current	3					
VLMR independent reference	independent code	current	3,381					
numDeriv / dense likelihood	independent code	current	33					
internal identity	internal	current	15					
runtime guard	internal	current	14					
internal check	internal	current	11		65			

Candidates not yet wired in

reference	kind	data	published numbers	checked
mclust bootstrap LRT (2016 paper; vignette)	R package	mclust::diabetes (two versions)	statistics 361.186445 / 114.703559 / 7.437806	verified
randomLCA (Beath 2017)	R package	randomLCA::myocardial	bootstrap LRT p = .001	verified
Masyn (2013) Table 25.12 and Stata lcstats	published	Reaven-Miller /10	adjusted LMR p-values	verified
Mplus Demo TECH11/TECH14 (JStats)	commercial program	mplus-lca-tech11/tech14-s1/s2; lca-boundary-tech11	TECH11/TECH14 for K = 2, 3	verified

enumerate_*() / compare_models()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
Mplus	commercial program	current	51	2				
mclust	R package	current	122	3	120			*
poLCA	R package	current	47	3	36			*
glca	R package	current	2					*
Kneedle hand-worked and reported values	independent code	current	7					
internal check	internal	current	30		87			
refusal parity	internal	current	16		16			

pool_imputations()

reference	kind	status	current checks	datasets	values	stale checks	files without result	shared
mitools	R package	current	163	2	2,148			
mice	R package	current	62	1	1,498			
mitml	R package	current	20	2	425			
mice pool.scalar	component check	current	52					
stats p.adjust	component check	current	51		204			
Rubin's rules written here	independent code	current	294		4,252			
published formula computed here	independent code	current	147		1,467			
base-R likelihood + numDeriv	independent code	current	16		454			
internal check	internal	current	165		300			
direct multilpa() call	internal	current	14		444			
internal identity	internal	current	8					

Inputs

input	run	package version	used
validation/matrix-refresh-2026-11-14/equivalence/report.csv	2026-11-14	1.1.22	current results
validation/matrix-refresh-2026-11-14/argument-sweep/results.csv	2026-11-14	1.1.22	current results
validation/release-fixes-2026-10-28/release/external-tests-consolidated.csv	2026-10-28	0.98.0	only for files without a current result
validation/release-fixes-2026-10-28/release/retained-references-results.csv	2026-10-28	0.98.0	only for files without a current result
validation/matrix-refresh-2026-10-28/equivalence-results.csv	2026-10-29	0.98.1	only for files without a current result
validation/release-0.96.1-2026-10-20/outer-final/cases.csv	2026-10-20	0.96.1	only for files without a current result
validation/matrix-refresh-2026-11-01/equivalence-results.csv	2026-11-01	1.1.9 features	only for files without a current result
validation/matrix-refresh-2026-11-14/equivalence-tests/equivalence-a-f-results.csv	2026-11-14	1.1.22	current results
validation/matrix-refresh-2026-11-14/equivalence-tests/equivalence-g-l-results.csv	2026-11-14	1.1.22	current results
validation/matrix-refresh-2026-11-14/equivalence-tests/equivalence-m-u-v-e-w-z-results.csv	2026-11-14	1.1.22	current results
validation/matrix-refresh-2026-11-14/equivalence-tests/equivalence-verification-results.csv	2026-11-14	1.1.22	current results
validation/matrix-refresh-2026-11-14/equivalence-tests/tests-equivalence-results.csv	2026-11-14	1.1.22	current results

Attribution rules: harness-functions.csv, harness-references.csv, test-attribution.csv, sweep-functions.csv and sweep-references.csv beside build.R. lca(): every indicator categorical, one level; multilca(): the same with groups; lpa(): every indicator Gaussian, one level; multilpa(): everything else, including count and ordinal indicators and tests that call the engine directly.